

Name of Faculty	:	Faculty of Engineering & Technology
Name of Program	:	Bachelor of Engineering
Course Code	:	5BIT04
Course Title	:	Internet of Thing
Type of Course	:	Professional Elective (PE)
Year of Introduction	:	2025-26

Prerequisite	:	C programming, Microprocessor, Networking
Course Objective	:	To equip students with a comprehensive understanding of the Internet of Things (IoT), including its architecture, components, and applications. Students will learn about IoT protocols, communication technologies, sensor networks, data processing, and cloud integration.
Course Outcomes	:	At the end of this course, students will be able to:
	CO1	Demonstrate the design and operation of IoT systems, including Sensors, Microcontrollers, their integration and the required software, while taking into account various application areas.
	CO2	Analyze the different IoT protocols, outlining their components and overall operation within IoT systems to ensure effective communication.
	CO3	Design an IoT system that leverages cloud computing and storage, while addressing security concerns.
	CO4	Utilize IoT technologies to automate and address various real life challenges across different application areas.
	CO5	Create the software components of an IoT system using Arduino or Raspberry Pi programming.

Teaching and Examination Scheme

Teaching Scheme (Contact Hours)			Credits	Examination Marks				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	SEE	CIA	SEE	CIA	
3	0	2	4	50	50	25	25	150

Legends: L-Lecture; T-Tutorial/Teacher Guided Theory Practice; P – Practical, C – Credit, SEE – Semester End Examination, CIA - Continuous Internal Assessment (It consists of Assignments/Seminars/Presentations/MCQ Tests, etc.))

Course Content

Unit No.	Topics	Teaching Hours	Weightage	Mapping with CO
1	Introduction to Internet of Things: IoT Definition, Characteristics of IoT, Key Components of IoT, IoT Challenges, Enabling Technologies, Applications of IoT.	9	10%	
2	Sensors and Actuators: Sensors, Types of Sensors, SharpIR Distance, PIR motion Sensor, LDR Sensor, Gyro Sensor, Ultrasonic Distance Sensor and DHT Sensor, Controlling Sensors. Actuators: Servo Motor, Solenoid and stepper motor, Need of Relay while using Actuators. Interfacing sensors with microcontrollers.	9	15%	
3	Communication Models and Protocols for IoT: IoT communication Models: Device-to-Device, Device-to-Cloud, Device-to-Gateway Messaging protocols: AMQP, MQTT, CoAP and XMPP, Transport Protocols: BLE and LiFi. Networking Protocols: WiFi, Zigbee 5G, NFC	9	12%	
4	Cloud for IoT: IoT and Cloud, Edge Computing vs Cloud Computing for IoT, Cloud Platforms for IoT(AWS IoT, Google Cloud IoT, Microsoft Azure IoT), Security in Cloud, Fog Computing	9	8%	
5	Embedded Systems for IoT: Introduction to Microcontrollers (Arduino, Raspberry Pi, ESP32, etc.), Programming Embedded Systems, Interfacing Sensors and Actuators with Microcontrollers, Power Management in IoT devices.	9	8%	
6	Programming with Arduino Uno Block Diagram of Arduino UNO, Data types & Built in Constants, Operators, Control statements and Loops, Functions and library functions, Serial Communication Functions	9	10%	
7	Communication Models and Protocols for IoT: IoT communication Models: Device-to-Device, Device-to-Cloud, Device-to-Gateway Messaging protocols: AMQP, MQTT, CoAP and XMPP, Transport Protocols: BLE and LiFi. Networking Protocols: WiFi, Zigbee 5G, NFC	9	15%	

8	IoT Challenges and Future Directions: Scalability issues in IoT, IoT sustainability and Green Technology, IoT Governance, Ethics and Policy, Future Trends in IoT	9	7%	
9	IoT Application Development Building IoT-based Applications: IoT Smart Homes, Healthcare Transportation, Agricultural System, Smart Parking	9	15%	

Suggested Distribution of Theory Marks Using Bloom's Taxonomy						
Level	Remembrance	Understanding	Application	Analyse	Evaluate	Create
Weightage	20	35	35	0	0	0

NOTE: This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Suggested List of Experiments/Tutorials

Sr. No.	Name of Experiment/Tutorial	Teaching Hours
1	Study of Arduino Uno or Raspberry Pi board technical specifications and pins.	2
2	Study of various Sensors and actuators for IoT applications.	2
3	Develop IoT based application using Arduino board or Raspberry Pi to blink the on-board LED at one second interval.	2
4	Develop IoT based application using Arduino board or Raspberry Pi to turn ON and OFF the 3 LEDs (Red, Green, Blue) based on the 3 Pushbutton switches.	2
5	Develop IoT based application using Arduino board or Raspberry Pi to implement automated traffic signal system using different colour LED.	2
6	Develop IoT based application using Arduino board or Raspberry Pi to read analog input from a potentiometer and control the brightness of an LED.	2
7	Develop IoT based application using Arduino board or Raspberry Pi from DHT sensor to measure temperature and humidity.	2
8	Develop IoT based application using Arduino board or Raspberry Pi and ultrasonic sensor to measure distance and display it on the Serial Monitor.	2
9	Develop IoT based application using Arduino board or Raspberry Pi and PIR sensor to detect presence/movement and turn on an LED automatically.	2
10	Develop an IoT based application using Arduino board or Raspberry Pi to control servo motors back and forth across 180 degrees.	2

4	Develop data designs using DFDs (data flow diagram) and E-R (entity-Relationship) diagram.	2
5	Develop a class diagram for selected project.	2
6	Develop Sequence diagram for selected project.	2
7	Evaluate size of the project using Function point metric for the assigned Project.	4
8	Estimate cost of the project using COCOMO (Constructive Cost Model) /COCOMO II approach for the assigned project.	4
9	Use flow chart and Gantt charts to track progress of the assigned Project. (Use Sprint burn down chart if agile model is selected).	2
10	Prepare various test case for selected project.	2

Major Equipment/ Instruments and Software Required

Sr. No.	Name of Major Equipment/ Instruments and Software
1	Computer with latest configuration of windows or UNIX OS
2	Arduino IDE

Suggested Learning Websites

Sr. No.	Name of Website
1	Online Simulator for Arduino: https://www.tinkercad.com/dashboard
2	Arduino tutorial: https://www.arduino.cc/reference/en/
3	Tutorials point online learning portal: https://www.tutorialspoint.com/internet_of_things/index.htm
4	Swayam MOOC online course for the introduction to internet of Things: https://onlinecourses.nptel.ac.in/noc24_cs35/preview

Reference Books

Sr. No.	Name of Reference Books
1	Internet of Things: Architecture and Design Principles by Rajkamal
2	IoT Fundamentals Networking Technologies, Protocols and Use Cases for Internet of Things by David Hanes, Gonzalo salgueiro
3	Designing the Internet of Things